

Flow-Based Construction for Auditable Redistricting Assignments

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Abstract

This draft presents T.17, a flow-based construction baseline for BISECT. The paper focuses on capacity and cost structure, deterministic fixture behavior, infeasibility witnesses, and audit-compatible summaries. The staged implementation is intentionally modest: it establishes the contract for flow-family constructors before larger solvers are introduced, and it does not claim global optimality, legal compliance, or political fairness.

1 Introduction

Flow formulations are a natural way to express capacity and assignment costs. T.17 adds a small deterministic flow-construction baseline that records validity and infeasibility in a form suitable for audit and later solver expansion.

2 Method

The staged method constructs assignments under population-capacity constraints and records whether a valid assignment exists for the fixture. When capacity is infeasible, the output carries an explicit witness rather than emitting an invalid final plan.

2.1 Reproducible Procedure

The constructor uses the graph and unit attributes supplied by the BISECT state pipeline. The baseline is feasibility- and audit-oriented. It is not yet the branch-and-cut or branch-and-price machinery planned for U.16 and U.17.

1. Create source-side supply from units and demand-side capacity targets for districts or construction bins.
2. Add eligible arcs according to the declared graph/profile and attach costs from the staged constructor configuration.

3. Enforce capacity bounds and extract assignments from selected flow.
4. Mark the result **valid** when assignment and validation checks pass.
5. Mark the result **infeasible** when the declared flow model cannot satisfy capacity, and **invalid** when validation fails after extraction.

Status	Meaning	Artifact behavior
valid	Model produces a validated assignment	Write final sidecars
infeasible	Model/profile cannot satisfy capacity	Write witness
invalid	Extracted assignment fails validation	Block final export

Table 1: Flow-construction status taxonomy. Infeasibility is relative to the declared model and profile.

3 Implementation

The implementation lives in `bisect-flow` and is exposed through `bisect state --structure flow-construction` and YAML `structure: flow-construction`. Summaries record status, edge cut, population deviation, parameter hash, and algorithm lineage.

4 Audit Artifacts

Flow-construction summaries become audit evidence through the RPLAN/RCTX sidecar path: assignments, capacity status, infeasibility witnesses where present, parameter hashes, and algorithm lineage are preserved for verifier checks. Passing the audit profile confirms the recorded contract, not global optimality or legal compliance.

The sidecar contract binds the exported assignment or witness to the declared flow model, costs, capacities, parameters, validation outcome, and algorithm lineage. Parameter hashes are intended to detect drift in the run configuration. An infeasibility witness is model-relative: it explains why the declared flow profile failed, not why no possible redistricting plan can exist.

5 Evaluation Plan

The current evidence layer includes path balance, infeasible capacity, determinism fixtures, and a benchmark-tier RPLAN package generated from the flow constructor on a 100-unit synthetic path. The next evidence layer should characterize real-data scale limits and compare flow construction with clustering, regionalization, and METIS baselines.

Claim	Evidence now	Missing evidence	Status
Path fixture balance	L0 path tests	None for fixtures	Current
Infeasible capacity witness	L0 infeasibility tests	Broader case sweep	Current
Deterministic output	L0 determinism tests	Larger graph sweep	Current
Audit lineage	L1 sidecar checks	Real export transcript	Current
Scale smoke	path100 benchmark package	Real-data benchmarks	Benchmark smoke

Table 2: Evidence ladder for T.17 claims. Feasibility, witness generation, and plan quality are separate evidence levels.

Constructor	Capacity behavior	Failure artifact	Current role
Flow construction	model capacity bounds	infeasibility witness	baseline
Capacity clustering	status/repair	witness or repair lineage	comparison
Regionalization	merge constraints	merge/status summary	comparison
Spectral	sweep balance	validation failure	comparison
METIS	external partitioning	manifest/validation result	comparison

Table 3: Failure-behavior comparison framing. Quantitative comparisons remain future work until commands and outputs are recorded.

6 Limitations

The current flow constructor is a baseline, not a mature min-cost-flow solver for every restricting instance. Capacity feasibility and district quality must be evaluated separately.

Flow feasibility is also model-relative. A successful flow assignment does not prove compactness, community preservation, partisan fairness, or legal compliance. An infeasibility witness explains failure under the declared graph, costs, capacities, and profile; it is not proof that no valid plan exists under another model.

7 Conclusion

T.17 brings flow construction into the BISECT algorithm family while preserving the shared final-plan audit path used across the T/U roadmap. Its role is a model-relative, auditable construction baseline; stronger exact optimization claims belong to the later U.16 and U.17 papers.

References

BISECT Project. T.17 flow-based construction implementation spec, 2026. BISECT internal specification.